CricketMind

A Multi-Modal Multi-Agent System for Cricket Analytics

Thesis Architecture Draft — Version 2 (14-Agent Roster)

1 The Core Concept

The Problem

Modern cricket analytics is heavily siloed. Visual analysis (a coach reviewing batting technique on video) and statistical analysis (an analyst studying strike rates and match-ups) are performed completely independently. No existing artificial intelligence system automatically bridges the gap between what a cricket shot *looks like* (kinematics) and what its historical outcome *is* (statistics). Furthermore, attempting to solve complex sports queries with a single Large Language Model (LLM) results in hallucination and loss of reasoning context.

A second, subtler problem motivates this revision. Cricket is not consumed by a single kind of user. A franchise director, a bowling coach, a physiotherapist and a broadcast producer all need answers derived from the *same* underlying delivery, but expressed in entirely different vocabularies, at different time horizons, and under different constraints. A monolithic "analytics assistant" collapses these into one averaged voice that serves none of them well.

The Solution: CricketMind

CricketMind is a multi-modal, multi-agent system (MAS) built on the LangGraph framework. Rather than relying on a single AI brain, the system delegates analytical work to a decentralised *team* of specialised agents, coordinated by a supervisor.

The system takes two inputs simultaneously:

1. A natural-language query, e.g. *"What is the tactical weakness of this batsman against spin?"*
2. A raw video clip of the batsman in action.

2 System Architecture Diagram

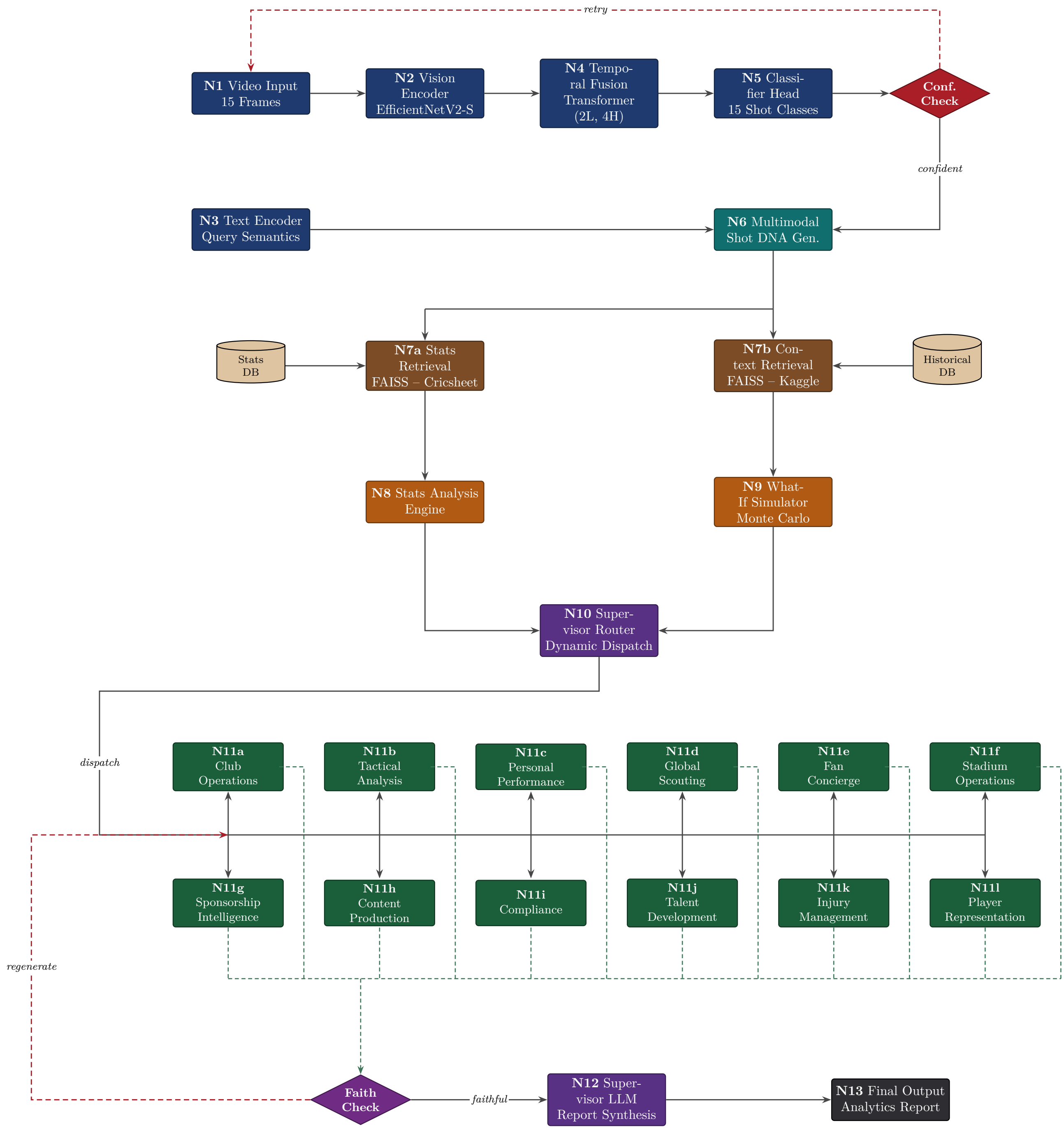

Figure 1: Full CricketMind LangGraph architecture (Version 2). Blue nodes perform visual and textual encoding; the teal node fuses both modalities into the Shot DNA record; brown nodes retrieve grounding evidence from the two FAISS indexes; orange nodes are the two shared analytical services that hold exclusive authority to emit numeric claims; green nodes are the twelve stakeholder-facing agents dispatched dynamically by the violet supervisor router. Red diamonds and red dashed edges are the two agentic correction loops; green dashed edges carry agent findings back to the faithfulness gate.

3 Node-by-Node Explanation

To make the flow concrete we trace a single user query end to end.

> **Running Example.** *"Compare Virat Kohli's Cover Drive vs. his Pull Shot against fast bowlers in the powerplay."*

3.1 Vision and Text Processing Layer (Blue Nodes)

N1 — Video Input (15 Frames).
The user uploads a clip of Kohli batting. The system uniformly samples **15 frames**, which is the temporal resolution the downstream backbone was trained on. Fifteen frames at roughly one second of footage is enough to capture the full stroke arc — backlift, stride, contact, follow-through — without paying for redundant near-duplicate frames.

N2 — Vision Encoder (EfficientNetV2-S).
Each of the 15 frames is passed through an EfficientNetV2-S backbone, producing a per-frame spatial embedding that encodes bat position, body posture, foot placement and ball location. The encoder is applied frame-independently; no temporal reasoning happens here.

N3 — Text Encoder (Query Semantics).
The natural-language query is embedded so that the system understands it must restrict attention to *fast bowlers* and the *powerplay*. Critically, N3 runs **in parallel** with the vision branch — it does not wait for classification — because its output is needed to parameterise retrieval, not to influence what the video contains.

N4 — Temporal Fusion (Temporal Transformer).
A lightweight transformer encoder — two layers, four attention heads, learnable positional embeddings and temporal mean pooling — consumes the ordered sequence of per-frame embeddings and models the stroke as a whole. Self-attention is precisely why this stage is not recurrent. In a gated recurrent unit, information from the pre-delivery stance reaches the follow-through only by surviving every intervening hidden state; under self-attention the first and last frames are a single attention hop apart. That direct long-range coupling is what separates strokes which differ not in their individual frames but in the ordering and velocity profile of the swing — the sweep and the reverse sweep being the canonical confusion pair for the recurrent baseline this replaces.

N5 — Classifier Head (15 Shot Classes).
A linear head projects the fused temporal representation onto the 15 CricShot10k shot classes and produces a probability distribution. *Example output:* 92% probability of **Cover Drive**.

3.2 Validation and Grounding (Red Diamond, Teal Node)

Confidence Check (Red Diamond).
If the top-1 classifier probability falls below a set threshold (e.g. 70%), the red dashed *retry* loop is triggered: the system re-enters N1 and attempts recovery by requesting an alternate camera angle, re-sampling a different 16-frame window, or applying test-time augmentation. The loop is bounded by a maximum retry count; on exhaustion the system degrades gracefully and reports low visual confidence rather than fabricating a label. This gate exists because *every* downstream claim is conditioned on the shot label — an incorrect label silently poisons all fourteen agents.

N6 — Multimodal Shot DNA Generation.
The accepted classification is fused with the query embedding from N3 and with continuous visual posture metrics (weight transfer, bat-face angle, stride length) into a single structured record we call the **Shot DNA**. This record is the sole interface between the perception half of the system and the analytics half: everything after N6 consumes Shot DNA and never touches raw pixels again.

3.3 Retrieval Layer (Brown Nodes)

N7a — Stats Retrieval (FAISS, Cricsheet).
Queries the ball-by-ball index for the precise conjunction implied by the Shot DNA and the query: *Kohli* + *Cover Drive* + *Powerplay* + *Pace*. This returns the narrow, high-precision evidence set.

N7b — Context Retrieval (FAISS, Kaggle).
Retrieves the broader historical envelope — Kohli's overall strike rate at this venue, his career trend against pace, comparable players. This is the low-precision, high-recall evidence set that prevents the system from over-fitting its narrative to a handful of deliveries.

3.4 Shared Analytical Services (Orange Nodes)

These two nodes are new in Version 2. They are not stakeholder agents; they have no audience of their own. They are **computational engines** that convert retrieved evidence into verified quantities, and they hold exclusive authority to emit a number. Every agent in the tier above consumes their output rather than performing arithmetic inside its own prompt.

N8 — Statistical Analysis Engine.
Performs the deterministic aggregation over the retrieved ball-by-ball records: strike rates, dismissal frequencies, control percentages, match-up splits. *Example output:* Kohli's strike rate is **142** on cover drives but only **112** on pull shots within the powerplay. Because this node is ordinary computation rather than generation, its results are reproducible and independently checkable — which is precisely what makes the faithfulness gate downstream meaningful.

N9 — What-If Simulator (Monte Carlo).
Answers counterfactual questions that no historical record can answer directly, by sampling forward from the retrieved distributions. *Example: "What if the bowler sends down six consecutive back-of-a-length deliveries?"* The Monte Carlo simulation returns a **40% chance** of Kohli being dismissed attempting the pull. This node is the system's only source of forward-looking, non-observational claims, and its outputs are always emitted with explicit uncertainty bounds.

3.5 Supervisor Router (Violet Node)

N10 — Supervisor Router (Dynamic Dispatch).
Inspects the Shot DNA, the query embedding and the analytical results, then activates **only the relevant subset** of the twelve stakeholder agents. A technique question wakes the Personal Performance and Talent Development agents; an in-match tactical question wakes Tactical Analysis; a workload question wakes Injury Management. Dispatching all twelve on every query would be both wasteful and harmful — it would bury the answer in irrelevant perspectives. Routing is the mechanism by which a 14-agent system stays legible.

3.6 The Twelve Stakeholder Agents (Green Nodes)

Each agent owns one stakeholder. It shares the same evidence as every other agent but differs in vocabulary, time horizon and output format.

N11a — Club Operations Agent (Franchises / Boards).
Squad budget modelling, IPL/BBL auction simulation, fixture scheduling, staff workflow coordination, travel logistics, contract-expiry alerts.

N11b — Tactical Analysis Agent (Coaches).
Ball-by-ball opposition profiling, pitch-condition-aware field placement, bowling strategy by match-up, batting order simulation, match scenario planning. On the running example this agent recommends bowling short to Kohli, reading directly off the pull-shot deficit computed by N8.

N11c — Personal Performance Agent (Players).
Shot-selection trend analysis, format-specific technique profiling, training load personalisation, milestone tracking, mental readiness integration. Here it notes that Kohli's visual weight transfer on the cover drive is excellent, reading the posture metrics carried in the Shot DNA.

N11d — Global Scouting Agent (Scouts).
Aggregates domestic and T20 league metrics across 20+ leagues, identifies emerging markets (Nepal, UAE, Afghanistan, USA), contract-cycle tracking, skill-gap-versus-franchise filtering.

N11e — Fan Concierge Agent (Fans).
Personalised highlight reels, live commentary preference engine, stadium wayfinding, ticket booking assistance, fantasy cricket advisory, merchandise recommendations.

N11f — Stadium Operations Agent (Stadiums).
Pitch curatorship with moisture and grass sensor integration, crowd flow modelling, concession demand forecasting, DRS review room scheduling, floodlight energy management.

N11g — Sponsorship Intelligence Agent (Sponsors).
Broadcast jersey and logo visibility scoring, audience demographic indexing, social media impression ROI, multilingual reach analysis, proposal auto-drafting.

N11h — Content Production Agent (Broadcasters).
Auto-generates multilingual commentary (Hindi, Bengali, Tamil, Sinhala, English), pitch map and wagon wheel overlays, highlights clipping, statistical ticker automation.

N11i — Compliance Agent (Governing Bodies).
ICC window conflict detection, player eligibility checks, franchise financial auditing, ACSU flag aggregation, bio-bubble protocol monitoring.

N11j — Talent Development Agent (Academies).
Personalised skill-acquisition roadmaps, age-group benchmark comparison (U-16 → U-19 → senior), drill recommendation, physical literacy progression tracking.

N11k — Injury Management Agent (Medical Teams).
Bowling workload tracking against overs-per-week thresholds, soft-tissue risk prediction, return-to-play timelines, session load versus fatigue modelling, hamstring and stress fracture flagging.

N11l — Player Representation Agent (Player Agents / Managers).
IPL auction valuation modelling, overseas league slot gap analysis, franchise squad roster scanning, market-demand heat maps by role and format.

3.7 Final Synthesis (Purple Diamond, Violet and Dark Nodes)

Faithfulness Check (Purple Diamond).
Before anything reaches the user, every numeric claim produced by the activated agents is checked against the values actually emitted by N8 and N9. If an agent asserts that Kohli's average is 999, or cites a strike rate the Statistical Analysis Engine never computed, the gate fails and the red dashed *regenerate* loop forces that agent to recompute against the real evidence. Version 2 makes this gate materially stronger than in Version 1: because N8 and N9 are the only sanctioned sources of numbers, the gate has a single ground-truth surface to compare against instead of having to adjudicate twelve independently generated arithmetic chains.

N12 — Supervisor LLM (Report Synthesis).
Takes the validated outputs arriving on the green dashed edges and composes them into one coherent, formatted report, resolving overlap between agents and ordering findings by relevance to the original query.

N13 — Final Output (Analytics Report).
The user receives the answer: *“Kohli’s cover drive is statistically superior against pace in the powerplay (SR 142 vs. 112). Our simulations show opposition bowlers should exploit his pull shot by bowling short…”*

4 Stakeholder Agent Roster

Stakeholder	Dedicated AI Agent	Primary Autonomous Capabilities
Franchises / Boards	**Club Operations Agent** (N11a)	Squad budget modelling, IPL/BBL auction simulation, fixture scheduling, staff workflow coordination, travel logistics, contract expiry alerts
Coaches	**Tactical Analysis Agent** (N11b)	Ball-by-ball opposition profiling, pitch-condition-aware field placement, bowling strategy recommendations by match-up, batting order simulator, match scenario planning
Players	**Personal Performance Agent** (N11c)	Shot-selection trend analysis, format-specific technique profiling, training load personalisation, milestone tracking, mental readiness integration
Scouts	**Global Scouting Agent** (N11d)	Aggregates domestic and T20 league metrics across 20+ leagues, identifies emerging markets, contract cycle tracking, skill-gap-vs-franchise filtering
Fans	**Fan Concierge Agent** (N11e)	Personalised highlight reels, live match commentary preference engine, stadium wayfinding, ticket booking assistant, fantasy cricket advisory, merchandise recommendations
Stadiums	**Stadium Operations Agent** (N11f)	Pitch curatorship AI (moisture/grass sensor integration), crowd flow modelling, food concession demand forecasting, DRS review room scheduling, floodlight energy management
Sponsors	**Sponsorship Intelligence Agent** (N11g)	Broadcast jersey/logo visibility scoring, audience demographic indexing, social media impression ROI, multilingual reach analysis, proposal auto-drafting
Broadcasters	**Content Production Agent** (N11h)	Auto-generates multilingual commentary (Hindi, Bengali, Tamil, Sinhala, English), pitch map / wagon wheel overlays, highlights clipping, statistical ticker automation
Governing Bodies	**Compliance Agent** (N11i)	ICC window conflict detection, player eligibility checks, franchise financial auditing, ACSU flag aggregation, bio-bubble protocol monitoring
Academies	**Talent Development Agent** (N11j)	Personalised skill-acquisition roadmaps, age-group benchmark comparison (U-16 → U-19 → senior), drill recommendation, physical literacy progression tracking
Medical Teams	**Injury Management Agent** (N11k)	Bowling workload tracking (overs-per-week thresholds), soft-tissue risk prediction, return-to-play timelines, session load vs. fatigue modelling, hamstring / stress fracture flagging
Player Agents / Managers	**Player Representation Agent** (N11l)	IPL auction valuation modelling, overseas league slot gap analysis, franchise squad roster scanning, market-demand heat maps by role and format
Shared analytical services (no dedicated stakeholder)		
All agents	**Statistical Analysis Engine** (N8)	Deterministic aggregation over retrieved ball-by-ball records: strike rates, dismissal frequencies, control percentages, match-up splits. Sole authority for observational numeric claims
All agents	**What-If Simulator** (N9)	Monte Carlo counterfactual simulation over retrieved distributions; forward-looking scenario probabilities with explicit uncertainty bounds

Table 1: The full 14-agent roster: twelve stakeholder-facing agents backed by two shared analytical services.

5 Control Loops

The architecture contains exactly two correction loops, both drawn as red dashed edges in Figure 1. Each guards a different failure mode.

The retry loop (perception failure).
Triggered by the Confidence Check when the classifier is unsure which shot it is looking at. Control returns to N1 for re-sampling or augmentation. This loop protects against *grounding* errors — reasoning correctly about the wrong shot.

The regenerate loop (generation failure).
Triggered by the Faithfulness Check when an agent's narrative contains a numeric claim unsupported by N8 or N9. Control returns to the dispatch bus and the offending agent is re-run against the verified evidence. This loop protects against *hallucination* — reasoning incorrectly about the right shot.

Both loops carry a bounded iteration counter. On exhaustion the system reports reduced confidence explicitly rather than emitting an unvalidated answer, which is a deliberate design choice: in a coaching or medical context a stated "insufficient evidence" is far more useful than a confident fabrication.